

Chapter 126 — Management and Considerations in Transgender and Nonbinary Individuals

INTRODUCTION

- **Transgender and gender-diverse (TGD)** is an umbrella term for a person whose **innate sense of being male or female (gender identity) differs from the sex assigned at birth**. Evidence of transgender people in society dates back to at least the 5th century BC across multiple cultures.
- **Gender expression** is how gender presents externally — clothing, hairstyle, behaviour, name, voice — which society may read as concordant or discordant with identity. **Pronouns may align with neither identity nor expression**.
- **Gender dysphoria** is the profound distress caused by the discrepancy between assigned sex and gender identity. It may be experienced by people identifying as men, women, transgender, agender, genderqueer, nonbinary or gender fluid.
- **Not all TGD individuals experience dysphoria**, and the need for medical intervention varies: **social transition alone suffices for some**, while others seek hormone treatment and/or **gender-affirming surgery (GAS)**. Transition is individual and may greatly alleviate dysphoria.
- TGD people face **marginalisation both in society and within health care**. Unmanaged dysphoria plus stigma produces clinically significant distress; reported stigma, prejudice, discrimination, harassment, abuse and violence generate **social, economic and legal marginalisation, poor mental and physical health outcomes, and death**.
- **Rising visibility has produced rising demand for high-quality gender-affirming care**, which general urologists will encounter.

TERMINOLOGY

- Language in transgender health is **in constant evolution**. Use of **culturally relevant, respectful and non-pathologising terminology** is critical to a safe and affirming relationship, **building rapport and reducing the mistrust many TGD people hold toward health systems**.

TABLE 126.1 TERMINOLOGY AND DEFINITIONS

Category	Terms and distinguishing content
Core constructs	Gender — a social construct varying with cultural norms, not a synonym for sex. Assigned sex — status as male, female or intersex on physical characteristics. Gender identity — the deeply felt, internal, intrinsic sense of one's own gender. Gender expression — the outward display of gender.
Assignment at birth	Assigned female at birth (AFAB) and assigned male at birth (AMAB) — assignment made on external genital appearance and/or genetics. Intersex — born with external genitalia not fitting binary definitions.
Concordance and discordance	Cisgender — identity congruent with sex assigned at birth. Transgender (TG) — umbrella term for identities or expressions not typically expected for the assigned sex. Gender diverse — identity or expression differing from that socially attributed to the assigned sex.
Directional terms	Transgender man — male identity, assigned female at birth. Transgender woman — female identity, assigned male at birth. Transmasculine / transfeminine — denote the direction of a gender-nonconforming or nonbinary person's gender, not a binary endpoint.
Nonbinary spectrum	

Category	Terms and distinguishing content
	Nonbinary — identifies as neither male nor female. Agender — identifies with no gender. Gender fluid — identity may change over time. Gender nonconforming / genderqueer — identity or expression outside traditional expectation; may be more complex, fluid or less clearly defined than that of a transgender person.
Diagnostic and distress terms	Gender dysphoria — the state of distress arising from the discrepancy between identity and assigned sex. Gender incongruence — the diagnostic term for marked and persistent incompatibility between identity and the gender expected from the assigned sex.
Process terms	Transition — the process of expressing one's gender identity; social, medical or surgical. Gender affirmation — being recognised in one's gender identity socially, medically, legally or behaviourally.
Orientation (a separate axis)	Sexual orientation — pattern of attraction and physiologic arousal; cannot be inferred from gender identity . Asexual — does not experience sexual attraction. Pansexual — attraction to people of all genders and sexes.
Pronouns	He/him/his masculine; she/her/hers feminine; they/them/theirs neutral; ze/hir/hirs, ze/zir/zirs, xe/xem/xyr used by some nonbinary or nonconforming people.
Historical, disfavoured	Transsexual — historical medical and public-health term for transgender people who have altered, or wish to alter, the body with hormones and/or surgery; debated and disfavoured in many settings .

From Byne et al., 2018; Coleman et al., 2022; Deutsch, 2016; Gonzalez et al., 2023; Hembree et al., 2017; Jha et al., 2022; Safer et al., 2019; Sahmoud et al., 2022; Smith, 2023; T'Sjoen et al., 2019.

- **What the table is for:** it keeps **four independent axes apart** — assigned sex, gender identity, gender expression and sexual orientation. Every terminology error is a collapse of one axis into another.
- **Dysphoria is a state of distress; incongruence is a diagnosis.** Not interchangeable.
- **Transmasculine and transfeminine denote direction, not destination**, and include people who do not seek a binary endpoint.

EPIDEMIOLOGY

- Approximately **25 million people worldwide** identify as TGD, including **over 1 million in the United States**.
- Prevalence rises steeply with how the question is asked: **lowest from health-systems studies (0.01–0.1%)**, higher from adult surveys (0.3–4.5%), **highest from child and adolescent surveys (2.5–8.4%)**, with a **predominance of transmasculine individuals**.
- All figures are **probable underestimates** — heterogeneous terminology, absence of large population-based studies, and under-reporting from fear of stigmatisation.

Concept — *Prevalence here is a function of who is asking and how. The gradient from health-systems data to adolescent surveys measures disclosure, not biology, so every figure is a floor rather than an estimate.*

EVALUATION AND DIAGNOSIS

Diagnostic and Statistical Manual of Mental Disorders, fifth edition (DSM-5), 2013

- Gender dysphoria requires **marked incongruence between experienced or expressed gender and sex assigned at birth**, present for **at least 6 months**.
- It must **not be solely the result of external prejudice or discrimination**, and must cause **clinically significant distress or impairment** in social, occupational or other important functioning.

International Classification of Diseases, 11th revision (ICD-11), released 2019

- **"Gender incongruence" replaced "gender identity disorder" of ICD-10.**
- It is **reclassified out of the mental disorders chapter into Conditions Related to Sexual Health** — the central examinable point, made to **reduce stigma and discrimination** while preserving access to care.
- Defined as **marked and persistent incongruence** between experienced gender and gender assigned at birth. Features: strong desire to be of a different gender; incongruence with primary and/or secondary sex characteristics; strong desire to possess the characteristics of the experienced gender; strong desire to be treated as a different gender.

Clinical Assessment

- Undertaken by a **qualified mental health professional, or a provider competent in the assessment of TGD individuals and gender-affirming care.**
- Content: detailed gender history (identity, expression, past and present distress, goals and desires); medical and mental health history; psychological evaluation **screening for depression, anxiety and conditions that may be associated with, exacerbate or mimic dysphoria.**
- Liaison with primary care or endocrinology **excludes underlying medical contributors** and establishes appropriateness for medical intervention.
- **Marked and sustained incongruence must be present, but severe distress is not a prerequisite for initiating gender-affirming therapy.**

Concept — *The diagnostic trend is from pathology to incongruence. ICD-11 retained a code so care remains accessible but moved it out of mental disorders; the threshold is marked and sustained incongruence, not the severity of suffering.*

WORLD PROFESSIONAL ASSOCIATION FOR TRANSGENDER HEALTH

- The **World Professional Association for Transgender Health (WPATH)** is an international, multidisciplinary body promoting evidence-based care, education and advocacy for TGD individuals.
- It publishes the **Standards of Care (SOC)**, covering psychological, medical, surgical and social aspects, used worldwide to inform practice. **Version 8, published September 2022**, is current.
- **TGD individuals treated in accordance with WPATH SOC show improved quality of life and overall well-being** — the bottom-line justification for the framework.

TABLE 126.2 RECOMMENDATIONS FOR HEALTH CARE PROFESSIONALS ASSESSING TRANSGENDER AND GENDER-DIVERSE (TGD) INDIVIDUALS FOR GENDER-AFFIRMING MEDICAL AND SURGICAL TREATMENT

- **Establishing the indication:** recommend treatment **only when gender incongruence is marked and sustained**; fulfil **diagnostic criteria** where a diagnosis is required locally to access care; **identify and exclude other possible causes** of apparent gender incongruence before initiation.
- **Optimising the patient:** assess **mental health conditions** and **physical health conditions** that could negatively affect outcome, with **risks and benefits discussed** before a treatment decision.
- **Consent and reproduction:** assess **capacity to consent to the specific physical treatment**; assess capacity to understand the **effect on reproduction** and **explore reproductive options before initiation**; **consider the role of social transition together with the individual.**
- **Process:** a **single opinion** from a professional competent in the assessment of TGD people suffices to initiate medical and surgical treatment.
- **Gonadectomy:** providers should consider a **minimum of 6 months of hormone therapy** appropriate to the person's gender goals before irreversible surgical intervention, **unless hormones are not clinically indicated.**

Modified from WPATH SOC Version 8.

- **What the table is for:** it marks the shift from **gatekeeping to structured, capacity-based, single-opinion assessment**, ensuring the individual understands options, risks and benefits before treatment.
- The **6-month hormone rule before gonadectomy is a consideration with an explicit exception**, not an absolute prerequisite.
- **Fertility counselling is mandated before initiation of treatment**, not merely before surgery.

KEY POINTS

- General urologists can expect to see and care for TGD patients.
- TGD identity is an umbrella term for a person whose innate sense of being male or female differs from the sex assigned at birth.
- WPATH publishes Standards of Care covering psychological, medical, surgical and social aspects of transgender health.
- TGD individuals receiving gender-affirming treatment in accordance with WPATH SOC experience improved quality of life and overall well-being.

Concept — *SOC 8 replaces thresholds the patient must clear with information the patient must receive. The surgeon's remaining hard obligations are capacity and fertility counselling, because those alone are irreversible.*

HORMONE TREATMENT

- **Gender-affirming hormone therapy (GAHT)** may be **medically necessary** to achieve changes consistent with gender identity or embodiment goals, as endorsed by **WPATH and the Endocrine Society**. Under medical supervision it is **safe and effective in reducing gender dysphoria**.
- Dosing principle: **target serum sex steroid levels aligned with gender identity, at physiologic levels**, by suppressing and/or supplementing hormones.
- **Eligibility (WPATH):** persistent well-documented gender dysphoria; capacity to make an informed decision and consent; **attainment of the age of majority in the person's country; controlled concomitant medical or psychiatric conditions**.
- **Fertility goals and preservation options are discussed before initiation.**

TABLE 126.3 GENDER-AFFIRMING HORMONE THERAPY REGIMENS

Direction	Class	Agent and route	Dose range
Transfeminine	Oestrogen	Oestradiol, oral	2.0–6.0 mg/day
Transfeminine	Oestrogen	Oestradiol, transdermal	0.025–0.2 mg/day; new patch every 3–5 days
Transfeminine	Oestrogen	Oestradiol valerate or cypionate, parenteral	5–30 mg IM every 2 weeks, or 2–10 mg IM weekly
Transfeminine	Antiandrogen	Spironolactone	100–300 mg/day
Transfeminine	Antiandrogen	Cyproterone acetate	25–50 mg/day
Transfeminine	Antiandrogen	Gonadotrophin-releasing hormone (GnRH) agonist	3.75 mg SC monthly, or 11.25 mg SC every 3 months
Transmasculine	Testosterone, parenteral	Testosterone enanthate or cypionate	100–200 mg SC every 2 weeks
Transmasculine		Testosterone undecanoate	1000 mg every 12 weeks

Direction	Class	Agent and route	Dose range
	Testosterone, parenteral		
Transmasculine	Testosterone, transdermal	Testosterone gel 1.6%	50–100 mg/day
Transmasculine	Testosterone, transdermal	Testosterone transdermal patch	2.5–7.5 mg/day

Modified from Hembree et al., 2017.

- **What the table is for:** the **structural asymmetry of the two regimens**. Transfeminine therapy is **two-component (oestrogen plus androgen blockade)**; transmasculine therapy is **single-component (testosterone alone)**, because physiologic-dose oestrogen does not suppress testosterone into the cisgender female range whereas testosterone suppresses the ovarian axis unaided.
- **Units are the trap:** oestradiol is monitored in **pg/mL**, testosterone in **ng/dL**.
- **Testosterone undecanoate** appears in the table but is **rarely used in the United States because of oil pulmonary embolism risk**. **Ethinyl oestradiol** is absent because it is not recommended.

Typical Regimens for Transfeminine Individuals

- **Goals:** **breast development, body fat redistribution, reduction of facial and body hair**. Achieved with **oestrogen plus an androgen-lowering agent**. Antiandrogens act by three mechanisms — **suppression of gonadotrophin secretion, inhibition of androgen biosynthetic enzymes, or direct androgen receptor antagonism**.

Oestrogen

- **Routes:** oral conjugated oestrogens, **oral 17 β -oestradiol**, transdermal 17 β -oestradiol, or intramuscular oestradiol.
- **Ethinyl oestradiol is not recommended** by the Endocrine Society: roughly **twentyfold increase in thromboembolic risk**, and it **cannot be directly monitored** on conjugated oestrogen assays.
- **Commonest United States regimen:** **oral 17 β -oestradiol 2–6 mg daily**, titrated to premenopausal cisgender female serum oestradiol of **100–200 pg/mL**.
- **Transdermal oestrogen** where venous thromboembolism (VTE) risk is higher — **age >45 years, or previous VTE**.

Antiandrogen

- Required because **physiologic-dose oestrogen monotherapy does not suppress total testosterone into the normal cisgender female range**.
- **Spironolactone is the commonest antiandrogen in the United States** — a potassium-sparing diuretic acting as a **moderate androgen receptor antagonist** and a **partial inhibitor of 17 α -hydroxylase/17,20-lyase**, so it also reduces testosterone synthesis.
- **Cyproterone acetate is standard of care in Europe**, per the European Network for Investigation of Gender Incongruence protocol.
- **Alternatives:** **GnRH agonists goserelin and leuprolide**, and the **nonsteroidal androgen receptor antagonist bicalutamide**.

5-Alpha Reductase Inhibitors

- **Finasteride and dutasteride block conversion of testosterone to dihydrotestosterone (DHT)**.
- **Less effective than spironolactone or cyproterone**, because they **neither reduce testosterone production nor block testosterone binding**.

- Useful for **spironolactone intolerance, those seeking partial feminisation, and persistent virilised features or hair loss after complete androgen blockade or orchidectomy.**

Progesterone

- **Role not well studied; data are limited.** Anecdotal reports of improved breast development, mood and sexual desire, set against **concern over thromboembolism and cerebrovascular accident** extrapolated from cisgender women.

Expected Changes, Risks and Monitoring

- **Physical changes within 3–12 months:** decreased sexual desire, decreased spontaneous erections, decreased facial and body hair, decreased sebum, increased skin softness, **breast tissue growth**, fat redistribution. **Long term: prostatic and testicular atrophy.**
- **Psychological:** decreased dysphoria, anxiety, depression and perceived stress, with improved quality of life. **Physiologic:** decreased systolic blood pressure, **decreased haematocrit**, increased triglycerides, **increased sex hormone-binding globulin**, decreased spermatogenesis.
- **Irreversibility is the examinable point: some breast development, and decline or absence of fertility, are irreversible.** Most other effects are at least partly reversible.
- **Risks parallel hormone replacement therapy (HRT) in cisgender people: thromboembolic disease, cardiovascular disease, hypertriglyceridaemia, liver dysfunction, osteoporosis, hormone-related cancers.** Risk arises from **supraphysiologic dosing, or from inadequate dosing failing to maintain normal physiology** — both ends of the dosing error.
- **Monitoring:** clinical review of weight, blood pressure and adverse effects; **sex steroid levels every 3 months during year 1, then once or twice yearly.** Where complete feminisation with suppression of male characteristics is sought, **testosterone <50 ng/dL and oestradiol 100–200 pg/mL**, checked 3-monthly.
- **On spironolactone: electrolytes and creatinine every 3 months in year 1, then annually.**
- **Prolactin at baseline, at least annually during transition, then every 2 years** — because of **prolactinoma risk after long-term, high-dose oestrogen.**
- **Bone density testing where osteoporotic risk factors exist:** previous fracture, family history, glucocorticoid use, prolonged hypogonadism. **Smoking cessation strongly encouraged** for the additive VTE risk.

Hormone Therapy Adjustment After Gender-Affirming Surgery

- **An endocrinologist or experienced clinician oversees postoperative care. After gonadectomy, hormone replacement and surveillance are necessary** to prevent the effects of chronic hormone deficiency, per WPATH SOC.
- **Oestrogen dose is reduced by 50–75% after gonadectomy, to a dose adequate to maintain bone density.**
- **Adequacy of the reduced dose is judged by luteinising hormone (LH) and follicle-stimulating hormone (FSH) levels.**
- **Antiandrogen therapy can be discontinued at this stage.**

Typical Regimens for Transmasculine Individuals

- Principles follow **HRT protocols with testosterone for hypogonadal cisgender males.**
- **Screen before initiation for contraindications: pregnancy, unstable coronary artery disease, untreated polycythaemia.**
- **History of oestrogen-sensitive malignancy warrants oncology referral** — testosterone aromatises to oestrogen.

- **Polycystic ovary syndrome (PCOS)** may require additional evaluation, given an elevated risk of side effects.
- **Effective contraception is essential** in those at risk of pregnancy, because of **testosterone teratogenicity**.

Testosterone and Adjuncts

- **Parenteral or transdermal routes**, targeting serum testosterone in the typical cisgender male range of 300–1000 ng/dL. Oral testosterone, axillary solutions, patches, nasal sprays, buccal tablets and pellets are less commonly used. Testosterone undecanoate is rarely employed in the United States because of oil pulmonary embolism risk.
- **GnRH analogues** may be used to **cease menses before initiating testosterone**, but are **less common in GAHT because of cost**.
- **Progesterone**: where **menses do not cease within a few months of starting testosterone**, consider a **progestational agent or endometrial ablation**.

Expected Changes, Risks and Monitoring

- **Early, from 1–2 months**: fat redistribution, **cessation of menses**, decreased fertility, increased sebum, acne, **clitoromegaly**, vaginal atrophy, increased sexual desire. **Later**: facial and body hair growth, increased muscularity, **deepening of the voice**. Dysphoria, anxiety, depression and perceived stress commonly decrease.
- **Irreversible**: clitoral growth, facial hair growth, voice change, male-pattern baldness.
- **Reversible with stated timelines**: cessation of menses reverses within **2–6 months**, vaginal atrophy within **3–6 months**, increased muscle mass within **6–12 months**. **Body hair growth and fat redistribution are variably reversible**.
- **Risks** extrapolated from hypogonadal cisgender men: **erythrocytosis, liver dysfunction, cardiovascular disease, sleep apnoea, excessive weight gain, cystic acne, hormone-related malignancy**. Long-term data in transmasculine individuals have been reassuring when therapy is used appropriately.
- **Monitoring**: evaluation every 3 months during year 1, then one to two times yearly, for virilisation and adverse reactions; **serum testosterone every 3 months until within the normal physiologic range; haematocrit at baseline and every 3 months in year 1, then one to two times yearly**, because of polycythaemia risk; routine weight, blood pressure and lipids.
- **Osteoporosis screening is targeted, not universal**: those who **discontinue testosterone, are non-adherent, or develop risk factors for bone loss**.

Hormone Therapy Adjustment After Gender-Affirming Surgery

- **After gonadectomy or hysterectomy, progesterone may be discontinued**.
- **No reduction in testosterone dosing is required after gonadectomy** — the direct contrast with the 50–75% oestrogen reduction.

Considerations in Nonbinary Individuals

- The principle is **individualised GAHT tailored to unique physical and functional goals**, not a binary endpoint: **defeminisation with or without masculinisation** for those assigned female at birth, or **demasculinisation with or without feminisation** for those assigned male at birth.
- **Assigned female at birth, seeking cessation of menses with partial masculinisation**: intrauterine device (IUD), progestins, GnRH analogues, endometrial ablation or hysterectomy, combined with testosterone supplementation and/or **nandrolone**, an anabolic steroid that **increases muscle mass without significantly increasing facial and body hair growth**.

- **Assigned male at birth, seeking full feminisation while preserving erectile function: oestrogen-only therapy**, or oestrogen with **low-dose antiandrogen** or a **5-alpha reductase inhibitor**.

Concept — *Transfeminine therapy is two-component and transmasculine therapy one-component, and every downstream difference follows from that asymmetry — the post-gonadectomy dose reduction, the withdrawal of antiandrogen, and the separate monitoring targets.*

UROLOGIC CONSIDERATIONS

- **Lower urinary tract symptoms (LUTS)** — frequency, urgency, nocturia, weak or hesitant stream, incomplete emptying, incontinence — are prevalent generally and also affect TGD patients.
- In TGD individuals LUTS arise from a defined set of contributors: **natal anatomy; hormonal therapy; gender-affirming surgery; pelvic floor dysfunction; and social, psychological and behavioural factors**. This list is the examinable framework.

Effect of Hormone Therapy

- Testosterone in transmasculine individuals **suppresses circulating oestrogen**, producing vaginal epithelial change **resembling that of postmenopausal cisgender women**.
- **Vaginal atrophy** follows: dryness, irritation, bleeding on penetration, dyspareunia, **urinary urgency and frequency**, with significant effect on quality of life.
- Management: **topical oestrogen to the vaginal mucosa as cream, ring or tablets**.

Effect of Genitourinary Surgery

- **After penile inversion vaginoplasty**, roughly a third of patients report altered voiding, a minority of whom are worse; **nocturia is the commonest complaint**.
- Possible mechanisms: **shortened urethra, prostatic urethral colonisation, vaginal dilation, underlying pelvic floor dysfunction**. Damage to neural innervation of bladder or sphincter is unlikely but possible.
- **Urinary stream divergence 5.6–33%**, attributed to **asymmetric labia, scarring, adhesive bands or residual corporal tissue**. Incontinence 4–16%, usually self-limiting. Urethral stricture 0–4%; meatal stenosis 1–6%; urethrovaginal fistula 0–4%, the last arising from unrecognised urethral injury.
- **After masculinising genitoplasty**, roughly a third of phalloplasty patients report altered voiding, a third of those worse. Post-phalloplasty dribbling arises from **pooling of urine at the junction between the native urethra and the pars fixa of the reconstructed urethra; compression at the base of the reconstructed scrotum** has been suggested to empty this space between voids.
- **Urethral stricture 2–63% after metoidioplasty with urethral lengthening; 7–58% after phalloplasty. Urethral fistula 5–50% after metoidioplasty with urethral lengthening; 18–78% after phalloplasty.**

Pelvic Floor, Behavioural Factors and Tucking

- Stress, anxiety, depression, trauma and dysphoria-related behaviour contribute to pelvic floor dysfunction and bothersome urinary symptoms.
- **Evaluation for pre-existing pelvic floor dysfunction before gender-affirming surgery is essential**, with timely treatment where identified.
- **Preoperative pelvic floor physical therapy before full-depth vaginoplasty** improves the postoperative experience of **neovaginal dilation**, by teaching relaxation of the pelvic floor musculature.
- **Tucking** is manual displacement of the testes upward into the inguinal canals, with penis and scrotal skin positioned between the legs and rearward toward the anus, maintained with tight underwear, tape, or a garment called a **gaff**; some continue during sleep.

- Consequences: **urinary stasis**, symptoms resembling **prostatitis, epididymo-orchitis or cystitis**, and **increased risk of lower urinary tract infection from prolonged compression of the meatus in proximity to the anus**.
- **Assessment of LUTS**: history, examination, and **urinalysis, uroflowmetry, ultrasonography for bladder emptying, cystoscopy, cross-sectional imaging and urodynamics** as clinically indicated.

Concept — *Reconstructing a urethra is far riskier than shortening one: masculinising genitoplasty lengthens the urethra and carries stricture and fistula rates an order of magnitude above vaginoplasty, which trades those for stream and emptying complaints. Non-surgical causes — atrophy, tucking, pelvic floor dysfunction — are excluded before LUTS are attributed to the operation.*

FERTILITY PRESERVATION

- Gender-affirming treatment carries consequences ranging from **impaired fertility to complete elimination of reproductive capacity**.
- **Counselling occurs before initiation of hormone therapy and/or surgery**, as recommended by **WPATH**, the American College of Obstetricians and Gynecologists (ACOG), the American Society for Reproductive Medicine (ASRM) and the Endocrine Society.
- Bottom line of the survey data: **a substantial minority of TGD individuals wish to have genetically related children** and actively consider preservation — relevant because many initiate GAHT in adolescence, when future fertility is not a salient concern.
- **Barriers**: cost, compounded by **lack of insurance coverage**; urgency to commence transition; difficulty with **future-oriented decision making**; **inadequate provider knowledge and bias** limiting the offer; and access difficulties. **Oocyte harvest and freezing costs several-fold more than sperm banking**, and **use of preserved gametes always requires assisted reproductive technology (ART)**, adding further expense. **The process itself may exacerbate dysphoria**.

Options for Transfeminine Individuals

- **Sperm cryopreservation is the most pursued option** — a well-established protocol, simple and affordable.
- Barrier: **ejaculation may itself evoke distress**, compounded by **antiandrogen effects on erection and ejaculation**. Alternatives: **vibratory stimulation, electro-ejaculation, or surgical retrieval from testis or epididymis**.
- **Prolonged GAHT progressively degrades semen quality** — increasing **oligozoospermia, asthenozoospermia, teratozoospermia and azoospermia**. GAHT should be held for at least 3 months before sperm collection.
- Cryopreserved sperm may be used for **in vitro fertilisation (IVF), intrauterine insemination (IUI), or intracytoplasmic sperm injection (ICSI) with IVF and embryo transfer**; alternatively, **embryo cryopreservation using fresh sperm and a donor oocyte**.
- **Testicular tissue cryopreservation is experimental but the sole option in prepubertal transfeminine individuals**. Tissue is obtained by **surgical biopsy**, which may be performed **concomitantly with gender-affirming surgery**, then **autotransplanted to testis, scrotum or ectopic subdermal sites** to allow maturation, followed by ART.

Options for Transmasculine Individuals

- **Embryo and/or oocyte cryopreservation are the most viable options**. Both require **controlled ovarian hyperstimulation (COH)** then **oocyte retrieval under sedation**, and entail considerable cost.

- **The examinable contrast:** embryo cryopreservation **requires fertilisation with donor or partner sperm at the time of freezing**; oocyte cryopreservation **preserves fertility without committing to a male partner or sperm donor**.
- **Ideally performed before commencing GAHT**, although **ovarian function may recover after testosterone exposure**, with recovery varying by **age and duration of testosterone use**.
- **Obstacles:** **temporary suspension of testosterone** causes undesired physical change and **resumption of endometrial bleeding**; and **pelvic examination, transvaginal ultrasonography and oocyte retrieval may themselves trigger emotional instability and distress**.
- **Ovarian tissue cryopreservation is the only option requiring neither cessation of GAHT nor controlled ovarian hyperstimulation**, and is the **sole option in prepubertal transmasculine individuals**. Tissue is obtained by **surgical biopsy, possibly concurrent with gender-affirming surgery**; on reimplantation **primordial follicles mature, at which point GAHT must be suspended**.
- **Fertility generally resumes soon after cessation of testosterone: menstruation returns within 3 to 6 months, and the majority conceive within 6 months. Conception on testosterone is possible, but testosterone must be discontinued when attempting to conceive, because of teratogenicity.**

Concept — *Gametes must be secured before hormones, because hormones degrade what surgery then removes. The only options surviving established GAHT are the experimental tissue-based ones; everything else needs a hormone holiday or a pre-treatment window that closes permanently at gonadectomy.*

GENITOURINARY CANCER SURVEILLANCE

- The field is **understudied, limited by data availability**, and the **long-term effect of GAHT on cancer risk remains largely unknown**.
- **TGD individuals who retain their natal reproductive organs remain at risk for the corresponding cancers** — the governing principle.
- **Risk drivers:** **stigma-associated smoking and excessive alcohol use**; higher prevalence of **human immunodeficiency virus (HIV)**, **hepatitis** and **human papillomavirus (HPV)** associated with high-risk sexual behaviours including commercial sex work and unprotected intercourse; and **discrimination, economic marginalisation and unmet care needs**, producing increased **AIDS-related, liver and anal cancers**.
- **Barriers to cancer care:** **lack of clinician training, insufficient cultural competence, limited TGD-specific screening guidelines, discrimination, and disparities in employment and insurance coverage**, with the net effect of **later-stage diagnosis and decreased survival**. TGD patients with cancer tend to be **younger, less likely to be non-Hispanic White, and less likely to hold private insurance** than cisgender patients.

Prostate Cancer Screening in Transfeminine Individuals

- **Incidence is rare** — a reported occurrence rate of **0.04%**.
- Despite lower incidence than in cisgender males, **disease may exhibit more aggressive characteristics**, potentially contributing to **increased mortality**.
- **Most cases occur in those who initiated hormone therapy after the age of 50, suggesting occult disease**.
- **No established guidelines exist for this population**. Current **United States Preventive Services Task Force (USPSTF)** and **American Urological Association (AUA)** advice is **shared decision making on prostate-specific antigen (PSA) testing for ages 55 to 69 years**.
- **PSA interpretation on GAHT is the key adjustment: an upper limit of normal of 1 ng/mL, or an increase from nadir of 0.3 ng/mL or greater**.

- **Examination follows the anatomy:** natal anatomy or **vulvoplasty → digital rectal examination; full-depth vaginoplasty → neovaginal examination**, the prostate lying anterior to the vaginal wall.
- **Biopsy may be transrectal, perineal or transneovaginal**, depending on anatomy. Treatment selection weighs **pathology, staging, comorbidity, and prior or future desires for gender-affirming surgery**.

Testicular Cancer Screening in Transfeminine Individuals

- **Sparsely documented** — a comprehensive literature review found only three case reports.
- **Routine screening beyond patient self-examination is not currently recommended.**

Cervical Cancer Screening in Transmasculine Individuals

- **All individuals with a cervix undergo routine screening**, per USPSTF and the American Society for Colposcopy and Cervical Pathology (ASCCP).
- **Ages 21–29:** cervical cytology alone every 3 years. **Ages 30–65:** cytology alone every 3 years, or high-risk HPV (hrHPV) testing alone every 5 years, or hrHPV with cytology (cotesting) every 5 years.
- **Continue until the patient no longer has a cervix, or until over 66 years with two consecutive negative tests.**
- **Added risk:** active HIV infection is **twice as prevalent as in cisgender males and ten times as prevalent as in cisgender females** in the transgender population, and **HIV increases the likelihood of persistent HPV infection**, which drives cervical cancer.
- **Making the examination tolerable is part of the clinical task:** offer screening in a **primary care setting not predominantly serving people assigned female at birth**; use the **smallest available speculum**; give the patient **agency over speculum insertion**; offer a **support person**.
- **In those who have taken testosterone, oestradiol 1% cream vaginally for 1 to 2 weeks beforehand reduces pain from testosterone-induced vaginal atrophy.**

Concept — *Screen the organ, not the gender. What is retained determines what is screened; what is exogenous determines how the result is interpreted, and the shifted PSA threshold is the model for that second half.*

LONG-TERM FOLLOW-UP OF TRANSGENDER AND GENDER-DIVERSE INDIVIDUALS

- **Routine care: age-, risk-factor- and organ-appropriate screening.** TGD individuals face elevated risk of cardiovascular disease, osteoporosis and certain malignancies. **Patients must be educated about their own anatomy and the recommended screenings**, particularly after surgery, because **comprehensive records may not be readily available to health care providers**.
- **Monitoring hormone levels and therapy:** routine measurement per society guidance, to **ensure therapeutic dosing while mitigating adverse events**.
- **Monitoring for complications after gender-affirming surgery:** close early follow-up for **functional and cosmetic outcomes**; in the longer term, **patient education on recognising signs and symptoms of complications** and encouragement to seek follow-up.
- **Mental health considerations:** persistent medical and mental health needs tied to gender identity may require **further medical intervention, mental health support or advocacy**; regular follow-up is the vehicle for addressing them.

KEY POINTS

- When provided under medical supervision, GAHT has been shown to be safe and effective in reducing gender dysphoria.

- Fertility goals and preservation options should be discussed before the initiation of GAHT.
- TGD individuals may experience LUTS related to natal anatomy, hormonal therapy, gender-affirming surgery, pelvic floor dysfunction, and social, psychological and behavioural factors.
- Organ-specific cancer screenings are a part of primary medical care for TGD individuals.

Concept — *The patient becomes the custodian of their own anatomy: records fragment across transitions and institutions, so education about retained organs and their screening obligations is itself a clinical intervention.*

CREATING A TRANSGENDER- AND NONBINARY-AFFIRMING CLINICAL ENVIRONMENT

Trauma-Informed Care

- Minority groups, including those with diverse sexual orientations, gender identities and expressions, face heightened risk of **bias, discrimination, harassment, violence, anxiety, depression, self-harm, suicide attempts, substance abuse and poorer overall health** than heterosexual and cisgender populations.
- The **Substance Abuse and Mental Health Services Administration (SAMHSA)** advocates a **trauma-informed approach** for marginalised communities.
- Trauma-informed care is **holistic**, acknowledging the effect of trauma and adverse experience — **physical, sexual or emotional abuse, neglect, or exposure to violence** — on physical, emotional and mental well-being.
- **Key principle: a safe, affirming and supportive environment, attuned to potential triggers and vulnerabilities.**
- **Safety** — a physical and emotional environment free from further harm: **affirming language, respect for chosen name and pronouns, assured confidentiality.**
- **SAMHSA's "Four Rs":** **Realise** the commonality and impact of trauma; **Recognise** the manifestations of trauma-related behaviours; **Respond** by incorporating trauma knowledge into policies and practice; **Resist retraumatisation** through conscious effort and appropriate language.
- **Trust** — transparent, honest and respectful interaction, with opportunity to share and affirm identity; **peer support from LGBTQ+ groups** encouraged for mutual aid and external validation.
- **Collaboration and empowerment** — partnership in all aspects of care, respect for autonomy and decisions, and provision of **information, choices and control.**
- **Cultural sensitivity** — understanding and respect for cultural and individual difference, with inclusive and affirming care.

Creating an Inclusive Practice

- Gender-affirming care requires a **multidisciplinary team**: primary care, mental health, endocrinology, plastic surgery, urology, otolaryngology, gynaecology, reproductive endocrinology and infertility, dermatology, aesthetics, speech therapy and social work.
- It begins with **acknowledging biases and avoiding assumptions about gender or sexual identity**. TGD individuals report **being denied care, verbal harassment, and delaying or avoiding preventive care because of discrimination.**
- **The first point of contact in the health care system is crucial.** All health care workers and frontline staff require education in **terminology, preferred pronouns and cultural competence.**

- **Ask about gender identity if not recorded in the electronic medical record (EMR)**, along with chosen pronouns and preferred name. **Intake forms should be inclusive** and capture details for non-cisgender patients.
- **A detailed organ inventory** allows individualisation of history, examination and preventive counselling as recommended by the USPSTF — the single most transferable tool in this section.
- **Physical examination** can be distressing and is conducted with trauma-informed practices: **a supportive presence of the patient's choice; explanation of the examination and its purpose; the patient's preferred position** — for example **lithotomy for transfeminine and standing for transmasculine individuals**; and **a warning before physical touch**.

KEY POINTS

- Key principles of trauma-informed care include the creation of a safe, affirming and supportive environment for TGD individuals, while being attuned to potential triggers and vulnerabilities.
- A transgender-inclusive urology practice requires a multidisciplinary approach, engagement across all levels of a health care system, education of health care workers, inclusive intake forms, and trauma-informed care practices during evaluations.

Concept — *Access is the rate-limiting step, not technique. Late-stage presentation is produced at the front desk, on the intake form and at the first examination, which is why the organ inventory and trauma-informed practice sit alongside hormone dosing as core competencies.*